Literature Status: Positive Entropy Forces Infinite Mean Dimension on the Space of Measures

Run `fa_banach_001` — lane 1

13 August 2026

Status: literature already answered (fully, and more strongly). The open question in arXiv:1105.0360 asks whether positive entropy of a continuous self-map forces positive topological mean dimension of its push-forward action on probability measures. The main theorem of Burguet and Shi, arXiv:2206.10508, proves that this induced mean dimension is in fact infinite for every positive-entropy compact dynamical system.

1 The source question

B. Kloeckner, *A generalization of Hausdorff dimension applied to Hilbert cubes and Wasserstein spaces*, arXiv:1105.0360, proves that if X is compact and $\varphi : X \to X$ has positive topological entropy, then the induced push-forward map $\varphi_\#$ on the Wasserstein space $\mathscr{W}_p(X)$ has positive *metric* mean dimension. Directly after Corollary 1.3, the paper asks [1]:

Source question. *Is the (topological) mean dimension of $\varphi_\#$ positive whenever φ has positive entropy? The source notes that determining this even for one map φ would be interesting.*

Here $\mathscr{W}_p(X)$ is the space of Borel probability measures on the compact metric space X, equipped with a Wasserstein metric.

2 The later theorem

D. Burguet and R. Shi, *Topological mean dimension of induced systems*, arXiv:2206.10508, consider the same induced system. They write $\mathcal{M}(X)$ for the Borel probability measures on X with the weak-* topology and $T_*\mu = \mu \circ T^{-1}$. Their main theorem states [2]:

Burguet–Shi main theorem. *For any topological system (X, T) with positive topological entropy, the induced system $(\mathcal{M}(X), T_*)$ has infinite topological mean dimension. Equivalently,*

$$h_{\text{top}}(T) > 0 \iff \text{mdim}(T_*) > 0 \iff \text{mdim}(T_*) = \infty.$$

For compact X, every finite-order Wasserstein metric induces the weak-* topology on the probability measures. Thus the topological dynamical systems denoted by $(\mathscr{W}_p(X), \varphi_\#)$ in the source and $(\mathcal{M}(X), T_*)$ in the later paper are the same after setting $T = \varphi$. The later theorem therefore answers the exact source question for every positive-entropy map, and strengthens "positive" to "infinite."

No new mathematical result is claimed in this packet.

3 Primary-source evidence

The following crop is from arXiv:1105.0360, PDF page 5. It contains Corollary 1.3 and the ensuing open question.

not to grow under countable unions. Let us give a more dynamical application that uses the intertwining property in a crucial way.

Corollary 1.3. — *If X is compact and $\varphi : X \to X$ is a continuous map with positive topological entropy, then $\varphi_\#$ has positive metric mean dimension. More precisely*

$$\operatorname{mdim}_M(\varphi_\#, \mathrm{W}_p) \geqslant p\frac{\mathrm{h_{top}}(\varphi)}{\log 2}.$$

Metric mean dimension is a metric invariant of dynamical systems that refines entropy for infinite-entropy ones, introduced by Lindenstrauss and Weiss [**LW00**] in link with mean dimension, a topological invariant. The definition of mdim_M is recalled in Section 6.3.

Note that the constant in Corollary 1.3 is not optimal in the case of multiplicative maps $\times d$ acting on the circle: in [**Klo10b**] we prove the lower bound $p(d-1)$ (instead of $p \log_2 d$ here).

It is a natural question to ask whether the (topological) mean dimension of $\varphi_\#$ is positive as soon as φ has positive entropy. To determine this at least for some map φ would be interesting.

1.2.2. Embedding Hilbert cubes. — Since embedding powers cannot be enough

The following crop is from arXiv:2206.10508, PDF page 1. It records the definition of the induced system and the main theorem.

DAVID BURGUET AND RUXI SHI

Abstract. For a topological system with positive topological entropy, we show that the induced transformation on the set of probability measures endowed with the weak-$*$ topology has infinite topological mean dimension. We also estimate the rate of divergence of the entropy with respect to the Wasserstein distance when the scale goes to zero.

1. Introduction

Let (X,T) be a topological dynamical system, i.e. X is a compact metrizable space and $T: X \to X$ is a continuous map. Let $\mathcal{M}(X)$ be the space of Borel probability measures on X endowed with the weak-$*$ topology. Then the push-forward map $T_*: \mu \mapsto \mu \circ T^{-1}$ is a continuous map on $\mathcal{M}(X)$. The topological system $(\mathcal{M}(X), T_*)$ is called the *induced sytem* of (X,T) on probability measures.

W. Bauer and K. Sigmund [BS75] have investigated which dynamical properties of $(\mathcal{M}(X), T_*)$ are inherited from (X,T). In particular they obsversed that $h_{top}(T_*)$ is infinite when $h_{top}(T)$ is positive. Later B. Weiss and E. Glasner [GW95] showed that if (X,T) has zero topogical entropy, then so does $(\mathcal{M}(X), T_*)$. Therefore we have the following equivalences :

$$h_{top}(T) > 0 \Leftrightarrow h_{top}(T_*) > 0 \Leftrightarrow h_{top}(T_*) = \infty.$$

M. Gromov [Gro99] has introduced a new topological invariant of dynamical systems called *(topological) mean dimension* as a dynamical analogue of topological covering dimension. In some sense the mean dimension $\mathrm{mdim}(T)$ estimates the number of parameters that we need to describe the system (X,T) per unity of time. A dynamical system of finite entropy or finite dimension always has zero mean dimension. The mean dimension of the shift on d-cubes $([0,1]^d)^{\mathbb{Z}}$ is equal to d ([LW00, Proposition 3.3]). Computing the mean dimension is in general difficult and it is only known for few systems ([Tsu19, BS21]). In the present paper we investigate the mean dimension of the induced system $(\mathcal{M}(X), T_*)$. Our main result states as below :

Main Theorem. *For any topological system (X,T) with positive topological entropy, the induced system $(\mathcal{M}(X), T_*)$ has infinite topological mean dimension. Therefore,*

$$h_{top}(T) > 0 \Leftrightarrow \mathrm{mdim}(T_*) > 0 \Leftrightarrow \mathrm{mdim}(T_*) = \infty.$$

Given a distance d on X, Lindenstrauss and Weiss [LW00] has introduced the associated metric mean dimension $\mathrm{mdim}(X, d, T)$ as the dynamical quantity corresponding to the

References

[1] B. Kloeckner, *A generalization of Hausdorff dimension applied to Hilbert cubes and Wasserstein spaces*, arXiv:1105.0360 (2011).

[2] D. Burguet and R. Shi, *Topological mean dimension of induced systems*, arXiv:2206.10508 (2022).